

GraphTheorySymbol:

A set of $\text{\LaTeX}$ commands to have symbols designed for
Graph Theory

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Abstract

This documentation is for $\text{\LaTeX}$ package GraphTheorySymbol. It provides symbols used in graph theory. The adjacency and incidence symbols were created by the author of this package.

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1 Introduction

This documentation is for $\text{\LaTeX}$ package GraphTheorySymbol. It provides symbols used in graph theory. The adjacency and incidence symbols were created by the author of this package.

This package is licenced under Lesser GNU General Public License version 3 or later (LGPLv3+). It means you are free do anything with it, but if you do improve the code of the symbols that we provide, please do share these improvements in a similar way. There is no restriction on other graph theory symbols/code, even if it would be a good fit for inclusion in this package. There is also no need to share in a similar way what you build with this package. The text of the licenses is available at:

<https://www.gnu.org/licenses/>.

The git repository of this source code is also available at:

<https://github.com/LLyaudet/GraphTheorySymbol/>.

Clone it with this URL:

[git@github.com:LLyaudet/GraphTheorySymbol.git](https://github.com:LLyaudet/GraphTheorySymbol.git).

This package is available on CTAN:

<https://ctan.org/pkg/graph-theory-symbol>.

This is our first $\text{\LaTeX}$ package. It may be that our macros are quite slow and that a document making extensive use of these macros will be slow to compile. If you know how to produce better $\text{\LaTeX}$ code, please submit a pull request on GitHub.

2 Usage

2.1 Memo or for the impatient

Add in your preamble: `\usepackage{graph-theory-symbol}`.

Long name	Short name	Result	In context
<code>\GTSadjacency/</code>	<code>\GTSa/</code>	$\circ$	$u \circ v$
<code>\GTSadjacencyRelation/</code>	<code>\GTSaR/</code>	$\circ$	$u \circ v$
<code>\GTSunadjacency/</code>	<code>\GTSu/</code>	$\circ$	$u \circ v$
<code>\GTSunadjacencyRelation/</code>	<code>\GTSuR/</code>	$\circ$	$u \circ v$
<code>\GTSdirectedAdjacencyUp/</code>	<code>\GTSdAU/</code>	$\circ$	$u \circ v$
<code>\GTSdirectedAdjacencyUpRelation/</code>	<code>\GTSdAUR/</code>	$\circ$	$u \circ v$
<code>\GTSnegatedDirectedAdjacencyUp/</code>	<code>\GTSnDAU/</code>	$\circ$	$u \circ v$
<code>\GTSnegatedDirectedAdjacencyUpRelation/</code>	<code>\GTSnDAUR/</code>	$\circ$	$u \circ v$
<code>\GTSdirectedAdjacencyDown/</code>	<code>\GTSdAD/</code>	$\circ$	$u \circ v$
<code>\GTSdirectedAdjacencyDownRelation/</code>	<code>\GTSdADR/</code>	$\circ$	$u \circ v$
<code>\GTSnegatedDirectedAdjacencyDown/</code>	<code>\GTSnDAD/</code>	$\circ$	$u \circ v$
<code>\GTSnegatedDirectedAdjacencyDownRelation/</code>	<code>\GTSnDADR/</code>	$\circ$	$u \circ v$
<code>\GTSincidence/</code>	<code>\GTSi/</code>	$\circ$	$v \circ e$
<code>\GTSincidenceRelation/</code>	<code>\GTSiR/</code>	$\circ$	$v \circ e$
<code>\GTSnegatedIncidence/</code>	<code>\GTSnI/</code>	$\circ$	$v \circ e$
<code>\GTSnegatedIncidenceRelation/</code>	<code>\GTSnIR/</code>	$\circ$	$v \circ e$
<code>\GTSoutIncidence/</code>	<code>\GTSoI/</code>	$\circ$	$v \circ e$
<code>\GTSoutIncidenceRelation/</code>	<code>\GTSoIR/</code>	$\circ$	$v \circ e$
<code>\GTSnegatedOutIncidence/</code>	<code>\GTSnOI/</code>	$\circ$	$v \circ e$
<code>\GTSnegatedOutIncidenceRelation/</code>	<code>\GTSnOIR/</code>	$\circ$	$v \circ e$
<code>\GTSinIncidence/</code>	<code>\GTSiI/</code>	$\circ$	$v \circ e$
<code>\GTSinIncidenceRelation/</code>	<code>\GTSiIR/</code>	$\circ$	$v \circ e$
<code>\GTSnegatedInIncidence/</code>	<code>\GTSnII/</code>	$\circ$	$v \circ e$
<code>\GTSnegatedInIncidenceRelation/</code>	<code>\GTSnIIR/</code>	$\circ$	$v \circ e$
Long name	Short name	Result	In context
<code>\GTSadjacencyMonochrome/</code>	<code>\GTSaM/</code>	$\circ$	$u \circ v$
<code>\GTSadjacencyMonochromeRelation/</code>	<code>\GTSaMR/</code>	$\circ$	$u \circ v$
<code>\GTSunadjacencyMonochrome/</code>	<code>\GTSuM/</code>	$\circ$	$u \circ v$
<code>\GTSunadjacencyMonochromeRelation/</code>	<code>\GTSuMR/</code>	$\circ$	$u \circ v$
<code>\GTSdirectedAdjacencyUpMonochrome/</code>	<code>\GTSdAUM/</code>	$\circ$	$u \circ v$
<code>\GTSdirectedAdjacencyUpMonochromeRelation/</code>	<code>\GTSdAUMR/</code>	$\circ$	$u \circ v$
<code>\GTSnegatedDirectedAdjacencyUpMonochrome/</code>	<code>\GTSnDAUM/</code>	$\circ$	$u \circ v$
<code>\GTSnegatedDirectedAdjacencyUpMonochromeRelation/</code>	<code>\GTSnDAUMR/</code>	$\circ$	$u \circ v$
<code>\GTSdirectedAdjacencyDownMonochrome/</code>	<code>\GTSdADM/</code>	$\circ$	$u \circ v$
<code>\GTSdirectedAdjacencyDownMonochromeRelation/</code>	<code>\GTSdADMR/</code>	$\circ$	$u \circ v$
<code>\GTSnegatedDirectedAdjacencyDownMonochrome/</code>	<code>\GTSnDADM/</code>	$\circ$	$u \circ v$
<code>\GTSnegatedDirectedAdjacencyDownMonochromeRelation/</code>	<code>\GTSnDADMR/</code>	$\circ$	$u \circ v$
<code>\GTSincidenceMonochrome/</code>	<code>\GTSiM/</code>	$\circ$	$v \circ e$
<code>\GTSincidenceMonochromeRelation/</code>	<code>\GTSiMR/</code>	$\circ$	$v \circ e$
<code>\GTSnegatedIncidenceMonochrome/</code>	<code>\GTSnIM/</code>	$\circ$	$v \circ e$
<code>\GTSnegatedIncidenceMonochromeRelation/</code>	<code>\GTSnIMR/</code>	$\circ$	$v \circ e$
<code>\GTSoutIncidenceMonochrome/</code>	<code>\GTSoIM/</code>	$\circ$	$v \circ e$
<code>\GTSoutIncidenceMonochromeRelation/</code>	<code>\GTSoIMR/</code>	$\circ$	$v \circ e$
<code>\GTSnegatedOutIncidenceMonochrome/</code>	<code>\GTSnOIM/</code>	$\circ$	$v \circ e$
<code>\GTSnegatedOutIncidenceMonochromeRelation/</code>	<code>\GTSnOIMR/</code>	$\circ$	$v \circ e$
<code>\GTSinIncidenceMonochrome/</code>	<code>\GTSiIM/</code>	$\circ$	$v \circ e$
<code>\GTSinIncidenceMonochromeRelation/</code>	<code>\GTSiIMR/</code>	$\circ$	$v \circ e$
<code>\GTSnegatedInIncidenceMonochrome/</code>	<code>\GTSnIIM/</code>	$\circ$	$v \circ e$
<code>\GTSnegatedInIncidenceMonochromeRelation/</code>	<code>\GTSnIIMR/</code>	$\circ$	$v \circ e$

\GTSsetMainColor{blue}			
Long name	Short name	Result	In context
\GTSadjacency/	\GTSa/		u v
\GTSadjacencyRelation/	\GTSaR/		u v
\GTSunadjacency/	\GTSu/		u v
\GTSunadjacencyRelation/	\GTSuR/		u v
\GTSdirectedAdjacencyUp/	\GTSdAU/		u v
\GTSdirectedAdjacencyUpRelation/	\GTSdAUR/		u v
\GTSnegatedDirectedAdjacencyUp/	\GTSnDAU/		u v
\GTSnegatedDirectedAdjacencyUpRelation/	\GTSnDAUR/		u v
\GTSdirectedAdjacencyDown/	\GTSdAD/		u v
\GTSdirectedAdjacencyDownRelation/	\GTSdADR/		u v
\GTSnegatedDirectedAdjacencyDown/	\GTSnDAD/		u v
\GTSnegatedDirectedAdjacencyDownRelation/	\GTSnDADR/		u v
\GTSincidence/	\GTSi/		v e
\GTSincidenceRelation/	\GTSiR/		v e
\GTSnegatedIncidence/	\GTSnI/		v e
\GTSnegatedIncidenceRelation/	\GTSnIR/		v e
\GTSoutIncidence/	\GTSOI/		v e
\GTSoutIncidenceRelation/	\GTSOIR/		v e
\GTSnegatedOutIncidence/	\GTSnOI/		v e
\GTSnegatedOutIncidenceRelation/	\GTSnOIR/		v e
\GTSinIncidence/	\GTSiI/		v e
\GTSinIncidenceRelation/	\GTSiIR/		v e
\GTSnegatedInIncidence/	\GTSnII/		v e
\GTSnegatedInIncidenceRelation/	\GTSnIIR/		v e

2.2 Detailed usage

First you need to add in your preamble: `\usepackage{graph-theory-symbol}`.

All symbols macros were designed with ease of use in mind: they are called with a trailing slash, they can be called in text mode or in math mode, a variant with relation spacing for math mode exists with suffix “relation”.

`\GTS@newcommand` This macro will enforce that the other macros are called with a trailing slash, avoiding problems with spaces after macro call.

Arguments:

`{\langle commandname \rangle}` The name of the command to create.

`{\langle commandcode \rangle}` The code of the command to create.

`\GTSsetMainColor` This macro will set the main color used in GraphTheorySymbol. This color has initial value “black”.

Argument:

`{\langle color \rangle}` The color that must be used by GraphTheorySymbol afterward.

`\GTSsetSecondColor` This macro will set the second color used in GraphTheorySymbol. This color has initial value “gray”.

Argument:

`{\langle color \rangle}` The color that must be used by GraphTheorySymbol afterward.

`\GTS@a` This macro will draw any of the adjacency symbols if the correct arguments are given. It is not expected to be used directly. See the macros derived from it and `\GTS@b`: `\GTSxadjacency`, `\GTSxadj`, and `\GTSxa`. Arguments:

`{\langle mainColor \rangle}` If non-empty, this color will be used as main color.

`{\langle secondColor \rangle}` If non-empty, this color will be used as second color.

`{\langle drawSlash \rangle}` Non-empty implies that a negation will be drawn.

`{\langle drawUpArrow \rangle}` Non-empty implies that an up-arrow will be drawn.

`{\langle drawDownArrow \rangle}` Non-empty implies that a down-arrow will be drawn.

`\GTS@b` This macro builds on top of `\GTS@a` to handle mathematical relation spacing if sixth argument is not empty. It is not expected to be used directly. See the macros derived from it and `\GTS@a`: `\GTSxadjacency`, `\GTSxadj`, and `\GTSxa`. Arguments:

`{\langle mainColor \rangle}` If non-empty, this color will be used as main color.

`{\langle secondColor \rangle}` If non-empty, this color will be used as second color.

`{\langle drawSlash \rangle}` Non-empty implies that a negation will be drawn.

`{\langle drawUpArrow \rangle}` Non-empty implies that an up-arrow will be drawn.

`{\langle drawDownArrow \rangle}` Non-empty implies that a down-arrow will be drawn.

`{\langle relation \rangle}` Non-empty implies that relation spacing will be added.

`\GTSxa` Versatile macros for adjacency that can be given options. Argument:

`\GTSxadj` `[\langle optionsList \rangle]` Sets the following options if not empty:

`\GTSxadjacency` `mainColor=color1` or `mainC=color1` or `mColor=color1` or `mC=color1`,
`secondColor=color2` or `secondC=color2` or `sColor=color2` or `sC=color2`,
`monochrome` or `mon` or `m` (equivalent to `mC=black,sC=black`),
`negated` or `not` or `n`,
`up` or `u`,
`down` or `d`,
`relation` or `rel` or `r`.

`\GTSadjacency` This macro will draw an adjacency symbol. A first drawing didn’t include the small dots in the middle of the two circles corresponding to the graph vertices. But it was too similar to “spoon” and `multimap` symbols, and could be confused with a graph with only two vertices and one edge. Hence, these dots were added and they have the nice property to remind of the classical textual notation of adjacency relation as `Adj(.,.)`.

`A.\GTSadjacency/A.` or `A.\GTSadj/A.` or `A.\GTSa/A.` will yield $A \overset{\circ}{\circ} A$.

`u \GTSadjacency/ v` or `u \GTSadj/ v` or `u \GTSa/ v` will yield $u \overset{\circ}{\circ} v$.

`$u \GTSadjacency/ v$` or `$u \GTSadj/ v$` or `$u \GTSa/ v$` will yield $u \overset{\circ}{\circ} v$.

`\GTSadjacencyRelation` This macro will draw an adjacency symbol.

`\GTSadjRel` `$u \GTSadjacencyRelation/v$` or `$u \GTSadjRel/v$` or `$u \GTSaR/v$` will yield $u \overset{\circ}{\circ} v$.

`\GTSaR` This macro will draw an unadjacency symbol.

`\GTSunadjacency` `A.\GTSunadjacency/A.` or `A.\GTSunadj/A.` or `A.\GTSu/A.` will yield $A \overset{\circ}{\circ} A$.

`\GTSunadj` `u \GTSunadjacency/ v` or `u \GTSunadj/ v` or `u \GTSu/ v` will yield $u \overset{\circ}{\circ} v$.

`\GTSu` `$u \GTSunadjacency/ v$` or `$u \GTSunadj/ v$` or `$u \GTSu/ v$` will yield $u \overset{\circ}{\circ} v$.

`\GTSunadjacencyRelation` This macro will draw an unadjacency symbol.

`\GTSunadjRel` `$u \GTSunadjacencyRelation/v$` or `$u \GTSunadjRel/v$` or `$u \GTSuR/v$` will yield $u \overset{\circ}{\circ} v$.

`\GTSuR`

`\GTSdirectedAdjacencyUp` This macro will draw a directed adjacency upward symbol. The operand on the left of the symbol should be considered the source of the arc; the operand on the right of the symbol should be considered the target of the arc.

`\GTSdirAdjUp` $A.\overset{\circ}{\text{A}}$. or $A.\overset{\circ}{\text{A}}$. will yield $A.\overset{\circ}{\text{A}}$.

`\GTSdAU` $u \overset{\circ}{\text{v}}$ or $u \overset{\circ}{\text{v}}$ will yield $u \overset{\circ}{\text{v}}$.

$\$u \overset{\circ}{\text{v}}$ or $\$u \overset{\circ}{\text{v}}$ will yield $u \overset{\circ}{\text{v}}$.

`\GTSdirectedAdjacencyUp-` This macro will draw a directed adjacency upward symbol with relation spacing in math mode.

`\GTSdirAdjUpRel` $\$u \overset{\circ}{\text{v}}$ or $\$u \overset{\circ}{\text{v}}$ or $\$u \overset{\circ}{\text{v}}$ will yield $u \overset{\circ}{\text{v}}$.

`\GTSdAUR` will yield $u \overset{\circ}{\text{v}}$.

`\GTSnegatedDirected-` This macro will draw a negated directed adjacency upward symbol.

`AdjacencyUp` $A.\overset{\circ}{\text{A}}$. or $A.\overset{\circ}{\text{A}}$. will yield $A.\overset{\circ}{\text{A}}$.

`\GTSnegDirAdjUp` will yield $A.\overset{\circ}{\text{A}}$.

`\GTSnDAU` $u \overset{\circ}{\text{v}}$ or $u \overset{\circ}{\text{v}}$ or $u \overset{\circ}{\text{v}}$ will yield $u \overset{\circ}{\text{v}}$.

$\$u \overset{\circ}{\text{v}}$ or $\$u \overset{\circ}{\text{v}}$ or $\$u \overset{\circ}{\text{v}}$ will yield $u \overset{\circ}{\text{v}}$.

`\GTSnegatedDirected-` This macro will draw a negated directed adjacency upward symbol with relation spacing in math mode.

`AdjacencyUpRelation` $\$u \overset{\circ}{\text{v}}$ or $\$u \overset{\circ}{\text{v}}$ or $\$u \overset{\circ}{\text{v}}$ will yield $u \overset{\circ}{\text{v}}$.

`\GTSnegDirAdjUpRel` $\$u \overset{\circ}{\text{v}}$ or $\$u \overset{\circ}{\text{v}}$ or $\$u \overset{\circ}{\text{v}}$ will yield $u \overset{\circ}{\text{v}}$.

`\GTSnDAUR` will yield $u \overset{\circ}{\text{v}}$.

`\GTSdirectedAdjacencyDown` This macro will draw a directed adjacency downward symbol. The operand on the left of the symbol should be considered the target of the arc; the operand on the right of the symbol should be considered the source of the arc.

`\GTSdirAdjDown` $A.\overset{\circ}{\text{A}}$. or $A.\overset{\circ}{\text{A}}$. will yield $A.\overset{\circ}{\text{A}}$.

$u \overset{\circ}{\text{v}}$ or $u \overset{\circ}{\text{v}}$ or $u \overset{\circ}{\text{v}}$ will yield $u \overset{\circ}{\text{v}}$.

$\$u \overset{\circ}{\text{v}}$ or $\$u \overset{\circ}{\text{v}}$ or $\$u \overset{\circ}{\text{v}}$ will yield $u \overset{\circ}{\text{v}}$.

`\GTSdAD` will yield $u \overset{\circ}{\text{v}}$.

`\GTSdirectedAdjacencyDown-` This macro will draw a directed adjacency downward symbol with relation spacing in math mode.

`Relation` $\$u \overset{\circ}{\text{v}}$ or $\$u \overset{\circ}{\text{v}}$ or $\$u \overset{\circ}{\text{v}}$ will yield $u \overset{\circ}{\text{v}}$.

`\GTSdirAdjDownRel` $\$u \overset{\circ}{\text{v}}$ or $\$u \overset{\circ}{\text{v}}$ or $\$u \overset{\circ}{\text{v}}$ will yield $u \overset{\circ}{\text{v}}$.

`\GTSdADR` will yield $u \overset{\circ}{\text{v}}$.

`\GTSnegatedDirected-` This macro will draw a negated directed adjacency downward symbol.

`AdjacencyDown` $A.\overset{\circ}{\text{A}}$. or $A.\overset{\circ}{\text{A}}$. will yield $A.\overset{\circ}{\text{A}}$.

`\GTSnegDirAdjDown` will yield $A.\overset{\circ}{\text{A}}$.

`\GTSnDAD` $u \overset{\circ}{\text{v}}$ or $u \overset{\circ}{\text{v}}$ or $u \overset{\circ}{\text{v}}$ will yield $u \overset{\circ}{\text{v}}$.

$\$u \overset{\circ}{\text{v}}$ or $\$u \overset{\circ}{\text{v}}$ or $\$u \overset{\circ}{\text{v}}$ will yield $u \overset{\circ}{\text{v}}$.

`\GTSnegatedDirected-` This macro will draw a negated directed adjacency downward symbol with relation spacing in math mode.

`AdjacencyDownRelation` $\$u \overset{\circ}{\text{v}}$ or $\$u \overset{\circ}{\text{v}}$ or $\$u \overset{\circ}{\text{v}}$ will yield $u \overset{\circ}{\text{v}}$.

`\GTSnegDirAdjDownRel` $\$u \overset{\circ}{\text{v}}$ or $\$u \overset{\circ}{\text{v}}$ or $\$u \overset{\circ}{\text{v}}$ will yield $u \overset{\circ}{\text{v}}$.

`\GTSnDADR` $\$u \overset{\circ}{\text{v}}$ will yield $u \overset{\circ}{\text{v}}$.

`\GTSincidence` This macro will draw an incidence symbol. The operand on the left of the symbol should be considered the vertex to which the edge/the operand on the right of the symbol is incident.

`\GTSi` $A.\overset{\circ}{\text{A}}$. or $A.\overset{\circ}{\text{A}}$. will yield $A.\overset{\circ}{\text{A}}$.

$v \overset{\circ}{\text{e}}$ or $v \overset{\circ}{\text{e}}$ will yield $v \overset{\circ}{\text{e}}$.

$\$v \overset{\circ}{\text{e}}$ or $\$v \overset{\circ}{\text{e}}$ will yield $v \overset{\circ}{\text{e}}$.

`\GTSincidenceRelation` This macro will draw an incidence symbol with relation spacing in math mode.

`\GTSiR` $\$v \overset{\circ}{\text{e}}$ or $\$v \overset{\circ}{\text{e}}$ will yield $v \overset{\circ}{\text{e}}$.

`\GTSnegatedIncidence` This macro will draw a negated incidence symbol.

`\GTSnI` $A.\overset{\circ}{\text{A}}$. or $A.\overset{\circ}{\text{A}}$. will yield $A.\overset{\circ}{\text{A}}$.

$v \overset{\circ}{\text{e}}$ or $v \overset{\circ}{\text{e}}$ will yield $v \overset{\circ}{\text{e}}$.

$\$v \overset{\circ}{\text{e}}$ or $\$v \overset{\circ}{\text{e}}$ will yield $v \overset{\circ}{\text{e}}$.

`\GTSnegatedIncidence-` This macro will draw a negated incidence symbol with relation spacing in math mode.

`Relation` $\$v \overset{\circ}{\text{e}}$ or $\$v \overset{\circ}{\text{e}}$ will yield $v \overset{\circ}{\text{e}}$.

`\GTSnIR` $\$v \overset{\circ}{\text{e}}$ will yield $v \overset{\circ}{\text{e}}$.

`\GTSoutIncidence` This macro will draw a directed incidence upward (out(ward) incidence) symbol. The

`\GTSOI`

operand on the left of the symbol should be considered the vertex to which the arc/the operand on the right of the symbol is incident.

$A.\backslash\text{GTSoutIncidence}/A.$ or $A.\backslash\text{GTSOI}/A.$ will yield $A.\overset{\circ}{\circ}A.$

$v\ \backslash\text{GTSoutIncidence}/\ e$ or $v\ \backslash\text{GTSOI}/\ e$ will yield $v\ \overset{\circ}{\circ}e.$

$\$v\ \backslash\text{GTSoutIncidence}/\ e\$$ or $\$v\ \backslash\text{GTSOI}/\ e\$$ will yield $v\overset{\circ}{\circ}e.$

$\backslash\text{GTSoutIncidenceRelation}$ This macro will draw a directed incidence upward (out(ward) incidence) symbol with
 $\backslash\text{GTSOIR}$ relation spacing in math mode.

$\$v\backslash\text{GTSoutIncidenceRelation}/e\$$ or $\$v\backslash\text{GTSOIR}/e\$$ will yield $v\ \overset{\circ}{\circ}e.$

$\backslash\text{GTSnegatedOutIncidence}$ This macro will draw a negated directed incidence upward (out(ward) incidence) symbol.

$\backslash\text{GTSnOI}$ $A.\backslash\text{GTSnegatedOutIncidence}/A.$ or $A.\backslash\text{GTSnOI}/A.$ will yield $A.\overset{\circ}{\circ}A.$

$v\ \backslash\text{GTSnegatedOutIncidence}/\ e$ or $v\ \backslash\text{GTSnOI}/\ e$ will yield $v\ \overset{\circ}{\circ}e.$

$\$v\ \backslash\text{GTSnegatedOutIncidence}/\ e\$$ or $\$v\ \backslash\text{GTSnOI}/\ e\$$ will yield $v\overset{\circ}{\circ}e.$

$\backslash\text{GTSnegatedOutIncidence-Relation}$ This macro will draw a negated directed incidence upward (out(ward) incidence) symbol with relation spacing in math mode.

$\backslash\text{GTSnOIR}$ $\$v\backslash\text{GTSnegatedOutIncidenceRelation}/e\$$ or $\$v\backslash\text{GTSnOIR}/e\$$ will yield $v\ \overset{\circ}{\circ}e.$

$\backslash\text{GTSinIncidence}$ This macro will draw a directed incidence downward (in(ward) incidence) symbol. The
 $\backslash\text{GTSiI}$ operand on the left of the symbol should be considered the vertex to which the arc/the operand on the right of the symbol is incident.

$A.\backslash\text{GTSinIncidence}/A.$ or $A.\backslash\text{GTSiI}/A.$ will yield $A.\overset{\circ}{\circ}A.$

$v\ \backslash\text{GTSinIncidence}/\ e$ or $v\ \backslash\text{GTSiI}/\ e$ will yield $v\ \overset{\circ}{\circ}e.$

$\$v\ \backslash\text{GTSinIncidence}/\ e\$$ or $\$v\ \backslash\text{GTSiI}/\ e\$$ will yield $v\overset{\circ}{\circ}e.$

$\backslash\text{GTSinIncidenceRelation}$ This macro will draw a directed incidence downward (in(ward) incidence) symbol with
 $\backslash\text{GTSiIR}$ relation spacing in math mode.

$\$v\backslash\text{GTSinIncidenceRelation}/e\$$ or $\$v\backslash\text{GTSiIR}/e\$$ will yield $v\ \overset{\circ}{\circ}e.$

$\backslash\text{GTSnegatedInIncidence}$ This macro will draw a negated directed incidence downward (in(ward) incidence) sym-
 $\backslash\text{GTSnII}$ bol.

$A.\backslash\text{GTSnegatedInIncidence}/A.$ or $A.\backslash\text{GTSnII}/A.$ will yield $A.\overset{\circ}{\circ}A.$

$v\ \backslash\text{GTSnegatedInIncidence}/\ e$ or $v\ \backslash\text{GTSnII}/\ e$ will yield $v\ \overset{\circ}{\circ}e.$

$\$v\ \backslash\text{GTSnegatedInIncidence}/\ e\$$ or $\$v\ \backslash\text{GTSnII}/\ e\$$ will yield $v\overset{\circ}{\circ}e.$

$\backslash\text{GTSnegatedInIncidence-Relation}$ This macro will draw a negated directed incidence downward (in(ward) incidence) symbol with relation spacing in math mode.

$\backslash\text{GTSnIIR}$ $\$v\backslash\text{GTSnegatedInIncidenceRelation}/e\$$ or $\$v\backslash\text{GTSnIIR}/e\$$ will yield $v\ \overset{\circ}{\circ}e.$

$\backslash\text{GTSadjacencyMonochrome}$ This macro will draw a monochrome adjacency symbol.

$\backslash\text{GTSaM}$ $A.\backslash\text{GTSadjacencyMonochrome}/A.$ or $A.\backslash\text{GTSaM}/A.$ will yield $A.\overset{\circ}{\circ}A.$

$u\ \backslash\text{GTSadjacencyMonochrome}/\ v$ or $u\ \backslash\text{GTSaM}/\ v$ will yield $u\ \overset{\circ}{\circ}v.$

$\$u\ \backslash\text{GTSadjacencyMonochrome}/\ v\$$ or $\$u\ \backslash\text{GTSaM}/\ v\$$ will yield $u\overset{\circ}{\circ}v.$

$\backslash\text{GTSadjacencyMonochrome-Relation}$ This macro will draw a monochrome adjacency symbol with relation spacing in math mode.

$\backslash\text{GTSaMR}$ $\$u\backslash\text{GTSadjacencyMonochromeRelation}/v\$$ or $\$u\backslash\text{GTSaMR}/v\$$ will yield $u\ \overset{\circ}{\circ}v.$

$\backslash\text{GTSunadjacencyMonochrome}$ This macro will draw a monochrome unadjacency symbol.

$\backslash\text{GTSuM}$ $A.\backslash\text{GTSunadjacencyMonochrome}/A.$ or $A.\backslash\text{GTSuM}/A.$ will yield $A.\overset{\circ}{\circ}A.$

$u\ \backslash\text{GTSunadjacencyMonochrome}/\ v$ or $u\ \backslash\text{GTSuM}/\ v$ will yield $u\ \overset{\circ}{\circ}v.$

$\$u\ \backslash\text{GTSunadjacencyMonochrome}/\ v\$$ or $\$u\ \backslash\text{GTSuM}/\ v\$$ will yield $u\overset{\circ}{\circ}v.$

$\backslash\text{GTSunadjacencyMonochrome-Relation}$ This macro will draw a monochrome unadjacency symbol with relation spacing in math mode.

$\backslash\text{GTSuMR}$ $\$u\backslash\text{GTSunadjacencyMonochromeRelation}/v\$$ or $\$u\backslash\text{GTSuMR}/v\$$ will yield $u\ \overset{\circ}{\circ}v.$

$\backslash\text{GTSdirectedAdjacencyUp-Monochrome}$ This macro will draw a monochrome directed adjacency upward symbol. The operand
 $\backslash\text{GTSdAUM}$ on the left of the symbol should be considered the source of the arc; the operand on the right of the symbol should be considered the target of the arc.

$A.\backslash\text{GTSdirectedAdjacencyUpMonochrome}/A.$ or $A.\backslash\text{GTSdAUM}/A.$ will yield $A.\overset{\circ}{\circ}A.$

$u\ \backslash\text{GTSdirectedAdjacencyUpMonochrome}/\ v$ or $u\ \backslash\text{GTSdAUM}/\ v$ will yield $u\ \overset{\circ}{\circ}v.$

$\$u\ \backslash\text{GTSdirectedAdjacencyUpMonochrome}/\ v\$$ or $\$u\ \backslash\text{GTSdAUM}/\ v\$$ will yield $u\overset{\circ}{\circ}v.$

$\backslash\text{GTSdirectedAdjacencyUp-MonochromeRelation}$ This macro will draw a monochrome directed adjacency upward symbol with relation spacing in math mode.

$\backslash\text{GTSdAUMR}$ $\$u\backslash\text{GTSdirectedAdjacencyUpMonochromeRelation}/v\$$ or $\$u\backslash\text{GTSdAUMR}/v\$$ will yield $u\ \overset{\circ}{\circ}v.$

$\backslash\text{GTSnegatedDirected-AdjacencyUpMonochrome}$ This macro will draw a monochrome negated directed adjacency upward symbol.

$\backslash\text{GTSnDAUM}$ $A.\backslash\text{GTSnegatedDirectedAdjacencyUpMonochrome}/A.$ or $A.\backslash\text{GTSnDAUM}/A.$ will yield $A.\overset{\circ}{\circ}A.$

$u\ \backslash\text{GTSnegatedDirectedAdjacencyUpMonochrome}/\ v$ or $u\ \backslash\text{GTSnDAUM}/\ v$ will yield $u\ \overset{\circ}{\circ}v.$

$\$u\ \backslash\text{GTSnegatedDirectedAdjacencyUpMonochrome}/\ v\$$ or $\$u\ \backslash\text{GTSnDAUM}/\ v\$$ will yield $u\overset{\circ}{\circ}v.$

`\GTSnegatedDirectedAdjacencyUpMonochromeRelation` This macro will draw a monochrome negated directed adjacency upward symbol with relation spacing in math mode. `\GTSnDAUMR/v$` or `$u\GTSnDAUMR/v$` will yield $u \overset{\circ}{\frown} v$.

`\GTSdirectedAdjacencyDownMonochrome` This macro will draw a monochrome directed adjacency downward symbol. The operand on the left of the symbol should be considered the target of the arc; the operand on the right of the symbol should be considered the source of the arc. `A.\GTSdirectedAdjacencyDownMonochrome/A.` or `A.\GTSdADM/A.` will yield $A \overset{\circ}{\smile} A$. `u \GTSdirectedAdjacencyDownMonochrome/ v` or `u \GTSdADM/ v` will yield $u \overset{\circ}{\smile} v$. `$u \GTSdirectedAdjacencyDownMonochrome/ v$` or `$u \GTSdADM/ v$` will yield $u \overset{\circ}{\smile} v$.

`\GTSdirectedAdjacencyDownMonochromeRelation` This macro will draw a monochrome directed adjacency downward symbol with relation spacing in math mode. `\GTSdADMR` `$u\GTSdirectedAdjacencyDownMonochromeRelation/v$` or `$u\GTSdADMR/v$` will yield $u \overset{\circ}{\smile} v$.

`\GTSnegatedDirectedAdjacencyDownMonochrome` This macro will draw a monochrome negated directed adjacency downward symbol. `A.\GTSnegatedDirectedAdjacencyDownMonochrome/A.` or `A.\GTSnDADM/A.` will yield $A \overset{\circ}{\frown} A$. `u \GTSnegatedDirectedAdjacencyDownMonochrome/ v` or `u \GTSnDADM/ v` will yield $u \overset{\circ}{\frown} v$. `$u \GTSnegatedDirectedAdjacencyDownMonochrome/ v$` or `$u \GTSnDADM/ v$` will yield $u \overset{\circ}{\frown} v$.

`\GTSnegatedDirectedAdjacencyDownMonochromeRelation` This macro will draw a monochrome negated directed adjacency downward symbol with relation spacing in math mode. `\GTSnDADMR` `$u\GTSnegatedDirectedAdjacencyDownMonochromeRelation/v$` or `$u\GTSnDADMR/v$` will yield $u \overset{\circ}{\frown} v$.

`\GTSincidenceMonochrome` This macro will draw a monochrome incidence symbol. The operand on the left of the symbol should be considered the vertex to which the edge/the operand on the right of the symbol is incident. `A.\GTSincidenceMonochrome/A.` or `A.\GTSiM/A.` will yield $A \downarrow A$. `v \GTSincidenceMonochrome/ e` or `v \GTSiM/ e` will yield $v \downarrow e$. `$v \GTSincidenceMonochrome/ e$` or `$v \GTSiM/ e$` will yield $v \downarrow e$.

`\GTSincidenceMonochromeRelation` This macro will draw a monochrome incidence symbol with relation spacing in math mode. `\GTSiMR` `$v\GTSincidenceMonochromeRelation/e$` or `$v\GTSiMR/e$` will yield $v \downarrow e$.

`\GTSnegatedIncidenceMonochrome` This macro will draw a monochrome negated incidence symbol. `A.\GTSnegatedIncidenceMonochrome/A.` or `A.\GTSnIM/A.` will yield $A \uparrow A$. `v \GTSnegatedIncidenceMonochrome/ e` or `v \GTSnIM/ e` will yield $v \uparrow e$. `$v \GTSnegatedIncidenceMonochrome/ e$` or `$v \GTSnIM/ e$` will yield $v \uparrow e$.

`\GTSnegatedIncidenceMonochromeRelation` This macro will draw a monochrome negated incidence symbol with relation spacing in math mode. `\GTSnIMR` `$v\GTSnegatedIncidenceMonochromeRelation/e$` or `$v\GTSnIMR/e$` will yield $v \uparrow e$.

`\GTSoutIncidenceMonochrome` This macro will draw a monochrome directed incidence upward (out(ward) incidence) symbol. The operand on the left of the symbol should be considered the vertex to which the arc/the operand on the right of the symbol is incident. `A.\GTSoutIncidenceMonochrome/A.` or `A.\GTSoIM/A.` will yield $A \uparrow A$. `v \GTSoutIncidenceMonochrome/ e` or `v \GTSoIM/ e` will yield $v \uparrow e$. `$v \GTSoutIncidenceMonochrome/ e$` or `$v \GTSoIM/ e$` will yield $v \uparrow e$.

`\GTSoutIncidenceMonochromeRelation` This macro will draw a monochrome directed incidence upward (out(ward) incidence) symbol with relation spacing in math mode. `\GTSoIMR` `$v\GTSoutIncidenceMonochromeRelation/e$` or `$v\GTSoIMR/e$` will yield $v \uparrow e$.

`\GTSnegatedOutIncidenceMonochrome` This macro will draw a monochrome negated directed incidence upward (out(ward) incidence) symbol. `A.\GTSnegatedOutIncidenceMonochrome/A.` or `A.\GTSnOIM/A.` will yield $A \downarrow A$. `v \GTSnegatedOutIncidenceMonochrome/ e` or `v \GTSnOIM/ e` will yield $v \downarrow e$. `$v \GTSnegatedOutIncidenceMonochrome/ e$` or `$v \GTSnOIM/ e$` will yield $v \downarrow e$.

`\GTSnegatedOutIncidenceMonochromeRelation` This macro will draw a monochrome negated directed incidence upward (out(ward) incidence) symbol with relation spacing in math mode. `\GTSnOIMR` `$v\GTSnegatedOutIncidenceMonochromeRelation/e$` or `$v\GTSnOIMR/e$` will yield $v \downarrow e$.

`\GTSinIncidenceMonochrome` This macro will draw a monochrome directed incidence downward (in(ward) incidence) symbol. The operand on the left of the symbol should be considered the vertex to which the arc/the operand on the right of the symbol is incident. `A.\GTSinIncidenceMonochrome/A.` or `A.\GTSiM/A.` will yield $A \downarrow A$.

`v \GTSinIncidenceMonochrome/ e` or `v \GTSiIM/ e` will yield $v \not\propto e$.
`\GTSinIncidenceMonochrome/ e$` or `\GTSiIM/ e$` will yield $v \not\propto e$.
`\GTSinIncidenceMonochrome-Relation` This macro will draw a monochrome directed incidence downward (in(ward) incidence) symbol with relation spacing in math mode.
`\GTSiIMR` `\GTSinIncidenceMonochromeRelation/e$` or `\GTSiIMR/e$` will yield $v \not\propto e$.
`\GTSnegatedInIncidenceMonochrome` This macro will draw a monochrome negated directed incidence downward (in(ward) incidence) symbol.
`\GTSnIIM` `A.\GTSnegatedInIncidenceMonochrome/A.` or `A.\GTSnIIM/A.` will yield $A \not\propto A$.
`v \GTSnegatedInIncidenceMonochrome/ e` or `v \GTSnIIM/ e` will yield $v \not\propto e$.
`\GTSnegatedInIncidenceMonochrome/ e$` or `\GTSnIIM/ e$` will yield $v \not\propto e$.
`\GTSnegatedInIncidenceMonochromeRelation` This macro will draw a monochrome negated directed incidence downward (in(ward) incidence) symbol with relation spacing in math mode.
`\GTSnIIMR` `\GTSnegatedInIncidenceMonochromeRelation/e$` or `\GTSnIIMR/e$` will yield $v \not\propto e$.

3 Implementation

```

1 <{*package>
2 \RequirePackage{amsmath}
3 \RequirePackage{amssymb}
4 \RequirePackage{graphicx}
5 \RequirePackage{keyval}
6 \RequirePackage{fdsymbol}
7 \RequirePackage{lua-unicode-math}
8 \RequirePackage{xcolor}
9
10 \def\GTS@mainColor{black}
11 \def\GTS@secondColor{gray}
12 \def\GTS@mainColorBackup{black}
13 \def\GTS@secondColorBackup{gray}
14 \def\GTS@drawSlash{}
15 \def\GTS@drawUpArrow{}
16 \def\GTS@drawDownArrow{}
17
\GTS@newcommand This macro will enforce that the other macros are called with a trailing slash, avoiding
problems with spaces after macro call.
Arguments:
{<commandname>} The name of the command to create.
{<commandcode>} The code of the command to create.
18 \@ifdefinable{\GTS@newcommand}{\def\GTS@newcommand#1#2{%
19   \ifdefinable{#1}{\def#1/{#2}}
20 }}
\GTSsetMainColor This macro will set the main color used in GraphTheorySymbol. This color has initial value
“black”.
Argument:
{<color>} The color that must be used by GraphTheorySymbol afterward.
21 \newcommand{\GTSsetMainColor}[1]{%
22 \edef\GTS@mainColorTemp{#1}%
23 \edef\GTS@mainColorBackup{\GTS@mainColor}%
24 \edef\GTS@mainColor{\GTS@mainColorTemp}%
25 }
\GTSsetSecondColor This macro will set the second color used in GraphTheorySymbol. This color has initial
value “gray”.
Argument:
{<color>} The color that must be used by GraphTheorySymbol afterward.
26 \newcommand{\GTSsetSecondColor}[1]{%
27 \edef\GTS@secondColorTemp{#1}%
28 \edef\GTS@secondColorBackup{\GTS@secondColor}%
29 \edef\GTS@secondColor{\GTS@secondColorTemp}%
30 }
```


`\GTS@a` This macro will draw any adjacency symbol given the correct parameters. Its use is purely technical. See the macros derived from it and `\GTS@b`: `\GTSxadjacency`, `\GTSxadj`, and `\GTSxa`.

```

31 \newcommand{\GTS@a}[5]{%
32 \if#1\empty\else%
33 \GTSsetMainColor{#1}%
34 \fi%
35 \if#2\empty\else%
36 \GTSsetSecondColor{#2}%
37 \fi%
38 \text{\ensuremath{%
39 \raisebox{-0.2ex}{\scalebox{1.5}[1.5]{%
40 %
41 \raisebox{0.7ex}{\scalebox{0.7}[0.7]{%
42 $\mathcolor{\GTS@secondColor}{\circ}$%
43 }}%
44 \hspace{-0.39ex}%
45 \raisebox{1.05ex}{\scalebox{0.1}[0.1]{%
46 $\mathcolor{\GTS@secondColor}{\bullet}$%
47 }}%
48 \hspace{-0.12ex}%
49 \raisebox{0.11ex}{\scalebox{0.1}[0.73]{%
50 $\mathcolor{\GTS@mainColor}{\smallblacksquare}$%
51 }}%
52 \hspace{-0.40ex}%
53 \raisebox{-0.3ex}{\scalebox{0.7}[0.7]{%
54 $\mathcolor{\GTS@secondColor}{\circ}$%
55 }}%
56 \hspace{-0.39ex}%
57 \raisebox{0.05ex}{\scalebox{0.1}[0.1]{%
58 $\mathcolor{\GTS@secondColor}{\bullet}$%
59 }}%
60 \hspace{0.27ex}%
if GTS@drawUpArrow empty else
61 \if#4\empty\else%
62 \hspace{-0.65ex}%
63 \raisebox{-0.18ex}{\scalebox{0.5}[0.5]{%
64 \textcolor{\GTS@mainColor}{\textasciicircum}%
65 }}%
66 \fi%
if GTS@drawDownArrow empty else
67 \if#5\empty\else%
68 \hspace{-0.65ex}%
69 \raisebox{1.37ex}{\scalebox{0.5}[-0.5]{%
70 \textcolor{\GTS@mainColor}{\textasciicircum}%
71 }}%
72 \fi%
if GTS@drawSlash empty else
73 \if#3\empty\else%
74 \hspace{-0.65ex}%
75 \raisebox{0.25ex}{\scalebox{0.6}[0.6]{%
76 \textcolor{black}{/}%
77 }}%
78 \fi%
79 %
80 }}%
81 }}%
82 \if#1\empty\else%
83 \GTSsetMainColor{\GTS@mainColorBackup}%
84 \fi%
85 \if#2\empty\else%
86 \GTSsetSecondColor{\GTS@secondColorBackup}%
87 \fi%
88 }
```

`\GTS@b` This macro will handle wrapping in mathematical relation spacing if needed. Its use is purely technical. See the macros derived from it and `\GTS@a`: `\GTSxadjacency`, `\GTSxadj`, and `\GTSxa`.

```

89 \newcommand{\GTS@b}[6]{%
90 \if#61%
91 \mathrel{\GTS@a{#1}{#2}{#3}{#4}{#5}}%
92 \else%
93 \GTS@a{#1}{#2}{#3}{#4}{#5}%
94 \fi%
95 }

```

`\GTSxa` Versatile macros for adjacency that can be given options:

`\GTSxadj` `mainColor=color1` or `mainC=color1` or `mColor=color1` or `mC=color1`,
`\GTSxadjacency` `secondColor=color2` or `secondC=color2` or `sColor=color2` or `sC=color2`,
`monochrome` or `mon` or `m`,
`negated` or `not` or `n`,
`up` or `u`,
`down` or `d`,
`relation` or `rel` or `r`.

```

96 \define@key{GTSxa}{mainColor}{\def\GTS@@mainColor{#1}}
97 \define@key{GTSxa}{mainC}{\def\GTS@@mainColor{#1}}
98 \define@key{GTSxa}{mColor}{\def\GTS@@mainColor{#1}}
99 \define@key{GTSxa}{mC}{\def\GTS@@mainColor{#1}}
100 \define@key{GTSxa}{secondColor}{\def\GTS@@secondColor{#1}}
101 \define@key{GTSxa}{secondC}{\def\GTS@@secondColor{#1}}
102 \define@key{GTSxa}{sColor}{\def\GTS@@secondColor{#1}}
103 \define@key{GTSxa}{sC}{\def\GTS@@secondColor{#1}}
104 \define@key{GTSxa}{monochrome}[1]%
105 {\def\GTS@@mainColor{black}\def\GTS@@secondColor{black}}
106 \define@key{GTSxa}{mon}[1]%
107 {\def\GTS@@mainColor{black}\def\GTS@@secondColor{black}}
108 \define@key{GTSxa}{m}[1]%
109 {\def\GTS@@mainColor{black}\def\GTS@@secondColor{black}}
110 \define@key{GTSxa}{negated}[1]{\def\GTS@@drawSlash{1}}
111 \define@key{GTSxa}{not}[1]{\def\GTS@@drawSlash{1}}
112 \define@key{GTSxa}{n}[1]{\def\GTS@@drawSlash{1}}
113 \define@key{GTSxa}{up}[1]{\def\GTS@@drawUpArrow{1}}
114 \define@key{GTSxa}{u}[1]{\def\GTS@@drawUpArrow{1}}
115 \define@key{GTSxa}{down}[1]{\def\GTS@@drawDownArrow{1}}
116 \define@key{GTSxa}{d}[1]{\def\GTS@@drawDownArrow{1}}
117 \define@key{GTSxa}{relation}[1]{\def\GTS@@relation{1}}
118 \define@key{GTSxa}{rel}[1]{\def\GTS@@relation{1}}
119 \define@key{GTSxa}{r}[1]{\def\GTS@@relation{1}}
120
121 \newcommand{\GTSxa}[1] [] {%
122 \def\GTS@@mainColor{}%
123 \def\GTS@@secondColor{}%
124 \def\GTS@@drawSlash{}%
125 \def\GTS@@drawUpArrow{}%
126 \def\GTS@@drawDownArrow{}%
127 \def\GTS@@relation{}%
128 \setkeys{GTSxa}{#1}%
129 \GTS@b{\GTS@@mainColor}{\GTS@@secondColor}{\GTS@@drawSlash}%
130 {\GTS@@drawUpArrow}{\GTS@@drawDownArrow}{\GTS@@relation}%
131 }
132
133 \newcommand{\GTSxadj}[1] [] {\GTSxa[#1]}
134
135 \newcommand{\GTSxadjacency}[1] [] {\GTSxa[#1]}

```

`\GTSadjacency` This macro will draw an adjacency symbol.

```

\GTSadj 136 \GTS@newcommand{\GTSadjacency}{\GTS@a{}{}{}{}}
\GTSa 137 \GTS@newcommand{\GTSadj}{\GTS@a{}{}{}{}}
138 \GTS@newcommand{\GTSa}{\GTS@a{}{}{}{}}

```

`\GTSadjacencyRelation` This macro will draw an adjacency symbol with relation spacing in math mode.

```

\GTSadjRel 139 \GTS@newcommand{\GTSadjacencyRelation}{\mathrel{\GTSa/}}
\GTSaR 140 \GTS@newcommand{\GTSadjRel}{\mathrel{\GTSa/}}
141 \GTS@newcommand{\GTSaR}{\mathrel{\GTSa/}}

```

`\GTSunadjacency` This macro will draw an unadjacency symbol.

```

\GTSunadj 142 \GTS@newcommand{\GTSunadjacency}{\GTS@a{}{}{1}{}{}}
\GTSu 143 \GTS@newcommand{\GTSunadj}{\GTS@a{}{}{1}{}{}}
144 \GTS@newcommand{\GTSu}{\GTS@a{}{}{1}{}{}}

```

`\GTSunadjacencyRelation` This macro will draw an unadjacency symbol with relation spacing in math mode.

```

\GTSunadjRel 145 \GTS@newcommand{\GTSunadjacencyRelation}{\mathrel{\GTSu/}}
\GTSuR 146 \GTS@newcommand{\GTSunadjRel}{\mathrel{\GTSu/}}
147 \GTS@newcommand{\GTSuR}{\mathrel{\GTSu/}}

```

`\GTSdirectedAdjacencyUp` This macro will draw a directed adjacency upward symbol. The operand on the left of the symbol should be considered the source of the arc; the operand on the right of the symbol should be considered the target of the arc.

```

148 \GTS@newcommand{\GTSdirectedAdjacencyUp}{\GTS@a{}{}{1}{}{}}
149 \GTS@newcommand{\GTSdirAdjUp}{\GTS@a{}{}{1}{}{}}
150 \GTS@newcommand{\GTSdAU}{\GTS@a{}{}{1}{}{}}

```

`\GTSdirectedAdjacencyUpRelation` This macro will draw a directed adjacency upward symbol with relation spacing in math mode.

```

\GTSdirAdjUpRel 151 \GTS@newcommand{\GTSdirectedAdjacencyUpRelation}{\mathrel{\GTSdAU/}}
\GTSdAUR 152 \GTS@newcommand{\GTSdirAdjUpRel}{\mathrel{\GTSdAU/}}
153 \GTS@newcommand{\GTSdAUR}{\mathrel{\GTSdAU/}}

```

`\GTSnegatedDirectedAdjacencyUp` This macro will draw a negated directed adjacency upward symbol.

```

154 \GTS@newcommand{\GTSnegatedDirectedAdjacencyUp}{\GTS@a{}{}{1}{1}{}{}}
\GTSnegDirAdjUp 155 \GTS@newcommand{\GTSnegDirAdjUp}{\GTS@a{}{}{1}{1}{}{}}
\GTSnDAU 156 \GTS@newcommand{\GTSnDAU}{\GTS@a{}{}{1}{1}{}{}}

```

`\GTSnegatedDirectedAdjacencyUpRelation` This macro will draw a negated directed adjacency upward symbol with relation spacing in math mode.

```

\GTSnegDirAdjUpRel 157 \GTS@newcommand{\GTSnegatedDirectedAdjacencyUpRelation}{\mathrel{\GTSnDAU/}}
\GTSnDAUR 158 \GTS@newcommand{\GTSnegDirAdjUpRel}{\mathrel{\GTSnDAU/}}
159 \GTS@newcommand{\GTSnDAUR}{\mathrel{\GTSnDAU/}}

```

`\GTSdirectedAdjacencyDown` This macro will draw a directed adjacency downward symbol. The operand on the left of the symbol should be considered the target of the arc; the operand on the right of the symbol should be considered the source of the arc.

```

160 \GTS@newcommand{\GTSdirectedAdjacencyDown}{\GTS@a{}{}{}{1}{}{}}
161 \GTS@newcommand{\GTSdirAdjDown}{\GTS@a{}{}{}{1}{}{}}
162 \GTS@newcommand{\GTSdAD}{\GTS@a{}{}{}{1}{}{}}

```

`\GTSdirectedAdjacencyDownRelation` This macro will draw a directed adjacency downward symbol with relation spacing in math mode.

```

\GTSdirAdjDownRel 163 \GTS@newcommand{\GTSdirectedAdjacencyDownRelation}{\mathrel{\GTSdAD/}}
\GTSdADR 164 \GTS@newcommand{\GTSdirAdjDownRel}{\mathrel{\GTSdAD/}}
165 \GTS@newcommand{\GTSdADR}{\mathrel{\GTSdAD/}}

```

`\GTSnegatedDirectedAdjacencyDown` This macro will draw a negated directed adjacency downward symbol.

```

166 \GTS@newcommand{\GTSnegatedDirectedAdjacencyDown}{\GTS@a{}{}{}{1}{}{1}{}{}}
\GTSnegDirAdjDown 167 \GTS@newcommand{\GTSnegDirAdjDown}{\GTS@a{}{}{}{1}{}{1}{}{}}
\GTSnDAD 168 \GTS@newcommand{\GTSnDAD}{\GTS@a{}{}{}{1}{}{1}{}{}}

```

`\GTSnegatedDirectedAdjacencyDownRelation` This macro will draw a negated directed adjacency downward symbol with relation spacing in math mode.

```

\GTSnegDirAdjDownRel 169 \GTS@newcommand{\GTSnegatedDirectedAdjacencyDownRelation}{\mathrel{\GTSnDAD/}}
\GTSnDADR 170 \GTS@newcommand{\GTSnegDirAdjDownRel}{\mathrel{\GTSnDAD/}}
171 \GTS@newcommand{\GTSnDADR}{\mathrel{\GTSnDAD/}}

```

`\GTSincidence` This macro will draw an incidence symbol. The operand on the left of the symbol should be considered the vertex to which the edge/the operand on the right of the symbol is incident.

```

172 \GTS@newcommand{\GTSincidence}{%
173 \text{\ensuremath{%
174 \raisebox{0ex}{\scalebox{1.8}[1.8]{%
175 %
176 \raisebox{-0.3ex}{\scalebox{0.7}[0.7]{%
177 $\mathcolor{\GTS@secondColor}{\circ}$%
178 }}%
179 \hspace{-0.39ex}%
180 \raisebox{0.05ex}{\scalebox{0.1}[0.1]{%
181 $\mathcolor{\GTS@secondColor}{\bullet}$%
182 }}%
183 \hspace{-0.12ex}%
184 \raisebox{0.11ex}{\scalebox{0.1}[0.73]{%
185 $\mathcolor{\GTS@secondColor}{\smallblacksquare}$%
186 }}%
187 \hspace{-0.18ex}%
188 \raisebox{0.15ex}{\scalebox{0.2}[0.2]{%
189 $\mathcolor{\GTS@mainColor}{\smallblacksquare}$%
190 }}%
191 \hspace{0.2ex}%
192 \if\GTS@drawUpArrow\empty\else%
193 \hspace{-0.65ex}%
194 \raisebox{-0.18ex}{\scalebox{0.5}[0.5]{%
195 \textcolor{\GTS@secondColor}{\textasciicircum}%
196 }}%
197 \fi%
198 \if\GTS@drawDownArrow\empty\else%
199 \hspace{-0.65ex}%
200 \raisebox{1.37ex}{\scalebox{0.5}[-0.5]{%
201 \textcolor{\GTS@secondColor}{\textasciicircum}%
202 }}%
203 \fi%
204 \if\GTS@drawSlash\empty\else%
205 \hspace{-0.65ex}%
206 \raisebox{0.1ex}{\scalebox{0.5}[0.5]{%
207 \textcolor{black}{/}%
208 }}%
209 \fi%
210 %
211 }}%
212 }}%
213 }
214
215 \GTS@newcommand{\GTSi}{\GTSincidence/}

```

`\GTSincidenceRelation` This macro will draw an incidence symbol with relation spacing in math mode.

```

\GTSiR 216 \GTS@newcommand{\GTSincidenceRelation}{\mathrel{\GTSincidence/}}
217 \GTS@newcommand{\GTSiR}{\GTSincidenceRelation/}

```

`\GTSnegatedIncidence` This macro will draw a negated incidence symbol.

```

\GTSnI 218 \GTS@newcommand{\GTSnegatedIncidence}{%
219 \def\GTS@drawSlash{1}%
220 \GTSincidence/%
221 \def\GTS@drawSlash{}%
222 }
223
224 \GTS@newcommand{\GTSnI}{\GTSnegatedIncidence/}

```

`\GTSnegatedIncidenceRelation` This macro will draw a negated incidence symbol with relation spacing in math mode.

```

\GTSnIR 225 \GTS@newcommand{\GTSnegatedIncidenceRelation}{%
226 \mathrel{\GTSnegatedIncidence/}%
227 }
228 \GTS@newcommand{\GTSnIR}{\GTSnegatedIncidenceRelation/}

```

`\GTSoutIncidence` This macro will draw a directed incidence upward (out(ward) incidence) symbol. The operand on the left of the symbol should be considered the vertex to which the arc/the operand on the right of the symbol is incident.

```

229 \GTS@newcommand{\GTSoutIncidence}{%
230 \def\GTS@drawUpArrow{1}%
231 \GTSincidence/%
232 \def\GTS@drawUpArrow{}%
233 }
234
235 \GTS@newcommand{\GTSoI}{\GTSoutIncidence/}

```

`\GTSoutIncidenceRelation` This macro will draw a directed incidence upward (out(ward) incidence) symbol with relation spacing in math mode.

```

236 \GTS@newcommand{\GTSoutIncidenceRelation}{%
237 \mathrel{\GTSoutIncidence/}%
238 }
239 \GTS@newcommand{\GTSoIR}{\GTSoutIncidenceRelation/}

```

`\GTSnegatedOutIncidence` This macro will draw a negated directed incidence upward (out(ward) incidence) symbol.

```

240 \GTS@newcommand{\GTSnegatedOutIncidence}{%
241 \def\GTS@drawSlash{1}%
242 \GTSoutIncidence/%
243 \def\GTS@drawSlash{}%
244 }
245
246 \GTS@newcommand{\GTSnOI}{\GTSnegatedOutIncidence/}

```

`\GTSnegatedOutIncidenceRelation` This macro will draw a negated directed incidence upward (out(ward) incidence) symbol with relation spacing in math mode.

```

247 \GTS@newcommand{\GTSnegatedOutIncidenceRelation}{%
248 \mathrel{\GTSnegatedOutIncidence/}%
249 }
250 \GTS@newcommand{\GTSnOIR}{\GTSnegatedOutIncidenceRelation/}

```

`\GTSinIncidence` This macro will draw a directed incidence downward (in(ward) incidence) symbol. The operand on the left of the symbol should be considered the vertex to which the arc/the operand on the right of the symbol is incident.

```

251 \GTS@newcommand{\GTSinIncidence}{%
252 \def\GTS@drawDownArrow{1}%
253 \GTSincidence/%
254 \def\GTS@drawDownArrow{}%
255 }
256
257 \GTS@newcommand{\GTSiI}{\GTSinIncidence/}

```

`\GTSinIncidenceRelation` This macro will draw a directed incidence downward (in(ward) incidence) symbol with relation spacing in math mode.

```

258 \GTS@newcommand{\GTSinIncidenceRelation}{%
259 \mathrel{\GTSinIncidence/}%
260 }
261 \GTS@newcommand{\GTSiIR}{\GTSinIncidenceRelation/}

```

`\GTSnegatedInIncidence` This macro will draw a negated directed incidence downward (in(ward) incidence) symbol.

```

262 \GTS@newcommand{\GTSnegatedInIncidence}{%
263 \def\GTS@drawSlash{1}%
264 \GTSinIncidence/%
265 \def\GTS@drawSlash{}%
266 }
267
268 \GTS@newcommand{\GTSnII}{\GTSnegatedInIncidence/}

```

`\GTSnegatedInIncidenceRelation` This macro will draw a negated directed incidence downward (in(ward) incidence) symbol with relation spacing in math mode.

```

269 \GTS@newcommand{\GTSnegatedInIncidenceRelation}{%

```

```

270 \mathrel{\GTSnegatedInIncidence/}%
271 }
272 \GTS@newcommand{\GTSnIIR}{\GTSnegatedInIncidenceRelation/}

\GTSadjacencyMonochrome This macro will draw a monochrome adjacency symbol.
\GTSaM 273 \GTS@newcommand{\GTSadjacencyMonochrome}{%
274 \GTSsetMainColor{black}%
275 \GTSsetSecondColor{black}%
276 \GTSadjacency/%
277 \GTSsetMainColor{\GTS@mainColorBackup}%
278 \GTSsetSecondColor{\GTS@secondColorBackup}%
279 }
280
281 \GTS@newcommand{\GTSaM}{\GTSadjacencyMonochrome/}

\GTSadjacencyMonochrome- This macro will draw a monochrome adjacency symbol with relation spacing in math mode.
Relation 282 \GTS@newcommand{\GTSadjacencyMonochromeRelation}{%
\GTSaMR 283 \mathrel{\GTSadjacencyMonochrome/}%
284 }
285 \GTS@newcommand{\GTSaMR}{\GTSadjacencyMonochromeRelation/}

\GTSunadjacencyMonochrome This macro will draw a monochrome unadjacency symbol.
\GTSuM 286 \GTS@newcommand{\GTSunadjacencyMonochrome}{%
287 \def\GTS@drawSlash{1}%
288 \GTSadjacencyMonochrome/%
289 \def\GTS@drawSlash{}%
290 }
291
292 \GTS@newcommand{\GTSuM}{\GTSunadjacencyMonochrome/}

\GTSunadjacencyMonochrome- This macro will draw a monochrome unadjacency symbol with relation spacing in math
Relation mode.
\GTSuMR 293 \GTS@newcommand{\GTSunadjacencyMonochromeRelation}{%
294 \mathrel{\GTSunadjacencyMonochrome/}%
295 }
296 \GTS@newcommand{\GTSuMR}{\GTSunadjacencyMonochromeRelation/}

\GTSdirectedAdjacencyUp- This macro will draw a monochrome directed adjacency upward symbol. The operand on
Monochrome the left of the symbol should be considered the source of the arc; the operand on the right
\GTSdAUM of the symbol should be considered the target of the arc.
297 \GTS@newcommand{\GTSdirectedAdjacencyUpMonochrome}{%
298 \def\GTS@drawUpArrow{1}%
299 \GTSadjacencyMonochrome/%
300 \def\GTS@drawUpArrow{}%
301 }
302
303 \GTS@newcommand{\GTSdAUM}{\GTSdirectedAdjacencyUpMonochrome/}

\GTSdirectedAdjacencyUp- This macro will draw a monochrome directed adjacency upward symbol with relation spacing
MonochromeRelation in math mode.
\GTSdAUMR 304 \GTS@newcommand{\GTSdirectedAdjacencyUpMonochromeRelation}{%
305 \mathrel{\GTSdirectedAdjacencyUpMonochrome/}%
306 }
307 \GTS@newcommand{\GTSdAUMR}{\GTSdirectedAdjacencyUpMonochromeRelation/}

\GTSnegatedDirected- This macro will draw a monochrome negated directed adjacency upward symbol.
AdjacencyUpMonochrome 308 \GTS@newcommand{\GTSnegatedDirectedAdjacencyUpMonochrome}{%
\GTSnDAUM 309 \def\GTS@drawSlash{1}%
310 \GTSdirectedAdjacencyUpMonochrome/%
311 \def\GTS@drawSlash{}%
312 }
313
314 \GTS@newcommand{\GTSnDAUM}{\GTSnegatedDirectedAdjacencyUpMonochrome/}

```

`\GTSnegatedDirectedAdjacencyUpMonochrome-` This macro will draw a monochrome negated directed adjacency upward symbol with relation spacing in math mode.

```

Relation 315 \GTS@newcommand{\GTSnegatedDirectedAdjacencyUpMonochromeRelation}{%
\GTSnDAUMR 316 \mathrel{\GTSnegatedDirectedAdjacencyUpMonochrome/}%
317 }
318 \GTS@newcommand{\GTSnDAUMR}{%
319 \GTSnegatedDirectedAdjacencyDownMonochromeRelation/%
320 }
```

`\GTSdirectedAdjacencyDownMonochrome` This macro will draw a monochrome directed adjacency downward symbol. The operand on the left of the symbol should be considered the target of the arc; the operand on the right of the symbol should be considered the source of the arc.

```

321 \GTS@newcommand{\GTSdirectedAdjacencyDownMonochrome}{%
322 \def\GTS@drawDownArrow{1}%
323 \GTSadjacencyMonochrome/%
324 \def\GTS@drawDownArrow{}%
325 }
326
327 \GTS@newcommand{\GTSdADM}{\GTSdirectedAdjacencyDownMonochrome/}
```

`\GTSdirectedAdjacencyDownMonochromeRelation` This macro will draw a monochrome directed adjacency downward symbol with relation spacing in math mode.

```

\GTSdADMR 328 \GTS@newcommand{\GTSdirectedAdjacencyDownMonochromeRelation}{%
329 \mathrel{\GTSdirectedAdjacencyDownMonochrome/}%
330 }
331 \GTS@newcommand{\GTSdADMR}{\GTSdirectedAdjacencyDownMonochromeRelation/}
```

`\GTSnegatedDirectedAdjacencyDownMonochrome` This macro will draw a monochrome negated directed adjacency downward symbol.

```

332 \GTS@newcommand{\GTSnegatedDirectedAdjacencyDownMonochrome}{%
\GTSnDADM 333 \def\GTS@drawSlash{1}%
334 \GTSdirectedAdjacencyDownMonochrome/%
335 \def\GTS@drawSlash{}%
336 }
337
338 \GTS@newcommand{\GTSnDADM}{\GTSnegatedDirectedAdjacencyDownMonochrome/}
```

`\GTSnegatedDirectedAdjacencyDownMonochromeRelation` This macro will draw a monochrome negated directed adjacency downward symbol with relation spacing in math mode.

```

Relation 339 \GTS@newcommand{\GTSnegatedDirectedAdjacencyDownMonochromeRelation}{%
\GTSnDADMR 340 \mathrel{\GTSnegatedDirectedAdjacencyDownMonochrome/}%
341 }
342 \GTS@newcommand{\GTSnDADMR}{%
343 \GTSnegatedDirectedAdjacencyDownMonochromeRelation/%
344 }
```

`\GTSincidenceMonochrome` This macro will draw a monochrome incidence symbol. The operand on the left of the symbol should be considered the vertex to which the edge/the operand on the right of the symbol is incident.

```

345 \GTS@newcommand{\GTSincidenceMonochrome}{%
346 \GTSsetMainColor{black}%
347 \GTSsetSecondColor{black}%
348 \GTSincidence/%
349 \GTSsetMainColor{\GTS@mainColorBackup}%
350 \GTSsetSecondColor{\GTS@secondColorBackup}%
351 }
352
353 \GTS@newcommand{\GTSiM}{\GTSincidenceMonochrome/}
```

`\GTSincidenceMonochromeRelation` This macro will draw a monochrome incidence symbol with relation spacing in math mode.

```

Relation 354 \GTS@newcommand{\GTSincidenceMonochromeRelation}{%
\GTSiMR 355 \mathrel{\GTSincidenceMonochrome/}%
356 }
357 \GTS@newcommand{\GTSiMR}{\GTSincidenceMonochromeRelation/}
```

`\GTSnegatedIncidence-` This macro will draw a monochrome negated incidence symbol.

`Monochrome` 358 `\GTS@newcommand{\GTSnegatedIncidenceMonochrome}{%`
`\GTSnIM` 359 `\def\GTS@drawSlash{1}%`
360 `\GTSincidenceMonochrome/%`
361 `\def\GTS@drawSlash{}%`
362 `}`
363
364 `\GTS@newcommand{\GTSnIM}{\GTSnegatedIncidenceMonochrome/}`

`\GTSnegatedIncidence-` This macro will draw a monochrome negated incidence symbol with relation spacing in math mode.
`MonochromeRelation`

`\GTSnIMR` 365 `\GTS@newcommand{\GTSnegatedIncidenceMonochromeRelation}{%`
366 `\mathrel{\GTSnegatedIncidenceMonochrome/}%`
367 `}`
368 `\GTS@newcommand{\GTSnIMR}{\GTSnegatedIncidenceMonochromeRelation/}`

`\GTSoutIncidenceMonochrome` This macro will draw a monochrome directed incidence upward (out(ward) incidence) symbol. The operand on the left of the symbol should be considered the vertex to which the arc/the operand on the right of the symbol is incident.

`\GTSoIM` 369 `\GTS@newcommand{\GTSoutIncidenceMonochrome}{%`
370 `\def\GTS@drawUpArrow{1}%`
371 `\GTSincidenceMonochrome/%`
372 `\def\GTS@drawUpArrow{}%`
373 `}`
374
375 `\GTS@newcommand{\GTSoIM}{\GTSoutIncidenceMonochrome/}`

`\GTSoutIncidenceMonochrome-` This macro will draw a monochrome directed incidence upward (out(ward) incidence) symbol with relation spacing in math mode.
`Relation`

`\GTSoIMR` 376 `\GTS@newcommand{\GTSoutIncidenceMonochromeRelation}{%`
377 `\mathrel{\GTSoutIncidenceMonochrome/}%`
378 `}`
379 `\GTS@newcommand{\GTSoIMR}{\GTSoutIncidenceMonochromeRelation/}`

`\GTSnegatedOutIncidence-` This macro will draw a monochrome negated directed incidence upward (out(ward) incidence) symbol.
`Monochrome`

`\GTSnOIM` 380 `\GTS@newcommand{\GTSnegatedOutIncidenceMonochrome}{%`
381 `\def\GTS@drawSlash{1}%`
382 `\GTSoutIncidenceMonochrome/%`
383 `\def\GTS@drawSlash{}%`
384 `}`
385
386 `\GTS@newcommand{\GTSnOIM}{\GTSnegatedOutIncidenceMonochrome/}`

`\GTSnegatedOutIncidence-` This macro will draw a monochrome negated directed incidence upward (out(ward) incidence) symbol with relation spacing in math mode.
`MonochromeRelation`

`\GTSnOIMR` 387 `\GTS@newcommand{\GTSnegatedOutIncidenceMonochromeRelation}{%`
388 `\mathrel{\GTSnegatedOutIncidenceMonochrome/}%`
389 `}`
390 `\GTS@newcommand{\GTSnOIMR}{\GTSnegatedOutIncidenceMonochromeRelation/}`

`\GTSinIncidenceMonochrome` This macro will draw a monochrome directed incidence downward (in(ward) incidence) symbol. The operand on the left of the symbol should be considered the vertex to which the arc/the operand on the right of the symbol is incident.

`\GTSiIM` 391 `\GTS@newcommand{\GTSinIncidenceMonochrome}{%`
392 `\def\GTS@drawDownArrow{1}%`
393 `\GTSincidenceMonochrome/%`
394 `\def\GTS@drawDownArrow{}%`
395 `}`
396
397 `\GTS@newcommand{\GTSiIM}{\GTSinIncidenceMonochrome/}`

`\GTSinIncidenceMonochrome-` This macro will draw a monochrome directed incidence downward (in(ward) incidence) symbol with relation spacing in math mode.
`Relation`

`\GTSiIMR`


```

398 \GTS@newcommand{\GTSinIncidenceMonochromeRelation}{%
399 \mathrel{\GTSinIncidenceMonochrome/}%
400 }
401 \GTS@newcommand{\GTSiIMR}{\GTSinIncidenceMonochrome/}

\GTSnegatedInIncidence- This macro will draw a monochrome negated directed incidence downward (in(ward) inci-
Monochrome dence) symbol.
\GTSnIIM 402 \GTS@newcommand{\GTSnegatedInIncidenceMonochrome}{%
403 \def\GTS@drawSlash{1}%
404 \GTSinIncidenceMonochrome/%
405 \def\GTS@drawSlash{}%
406 }
407
408 \GTS@newcommand{\GTSnIIMR}{\GTSnegatedInIncidenceMonochrome/}

\GTSnegatedInIncidence- This macro will draw a monochrome negated directed incidence downward (in(ward) inci-
MonochromeRelation dence) symbol with relation spacing in math mode.
\GTSnIIMR 409 \GTS@newcommand{\GTSnegatedInIncidenceMonochromeRelation}{%
410 \mathrel{\GTSnegatedInIncidenceMonochrome/}%
411 }
412 \GTS@newcommand{\GTSnIIMR}{\GTSnegatedInIncidenceMonochromeRelation/}

413 \endinput
414 \</package>

```

4 Acknowledgments

Thank You God! Thank You Father! Thank You Jesus! Thank You Holy-Spirit!

5 Support

This package is gratis and free. But if you can star it on GitHub, it is a nice encouragement.

6 Change History

v0.1.0 – 2026-08-01	v1.1.1 – 2026-08-30
General: Initial version 1	General: Small corrections and improvements 1
v1.0.0 – 2026-08-26	v1.2.0 – 2026-09-06
General: First public release 1	General: CTAN URL, new macros for same set of symbols 1
v1.1.0 – 2026-08-29	
General: Added a memo 1	

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